

Tribhuvan University  
Faculty of Humanities & Social Sciences  
OFFICE OF THE DEAN

2025

## Bachelor in Computer Applications

Course Title: **Network Programming**

Code No: **CACS355** Semester: **VI**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

*Candidates are required to answer the questions in their own words as far as practicable.*

**Note:** Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

### Group B

2. Describe the scope and applications of Java Socket and Node.js in network programming platforms.
3. Explain the purpose of the `NetworkInterface` class and describe its factory and getter methods with examples.
4. Explain the process of opening a URL connection and reading data from the server in Java.
5. Write a Program to Send and Receive a Message using Connectionless Sockets.
6. What are the fundamental components of JAVA NIO? Difference between Blocking and Non-blocking I/O.
7. Explain the RMI architecture with the detail role.
8. Write short notes. (Any two) a) `ProxySelector` b) `JSSE` c) Web Cache

### Group C

*Attempt any TWO questions [2x10=20]*

9. a) What is Cookie? Explain about `CookieManager` and `HttpCookie` classes. b) Discuss the purpose and use of HTTP response codes in URL connections. Provide examples of common response codes.
10. Explain how a TCP echo server using thread created with snipped code and also give their advantages.
11. What is Traceroute? Write down the working principle. Explain the Trace Route program with sample code.